BCI Homework 3

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I used linear regression to predict arm movement variables from neural activity. The experiments show how training size, prediction target, PCA, proprioception, and neural history affect generalization. High training performance alone did not guarantee accurate test predictions.

Fold 10 was held out from fitting throughout. Tasks 3, 3.1, 4, and 6 used training folds 1–2. Unless varied, the model used 20 neural-history bins, no PCA, and no proprioception. FVAF measures prediction quality: 1 is perfect, 0 matches the evaluated set's mean-prediction baseline, and negative values are worse than that baseline.

Task 1 Basics and initialization

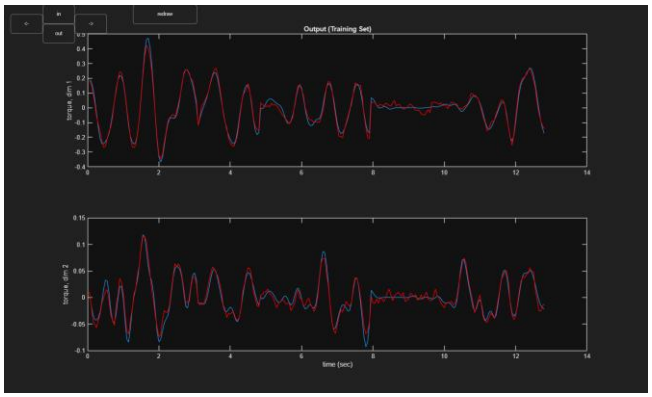
Torque output	Training FVAF	Test FVAF
Shoulder	0.9781	0.2607
Elbow	0.9400	−0.6840

Why does training performance differ from test performance?

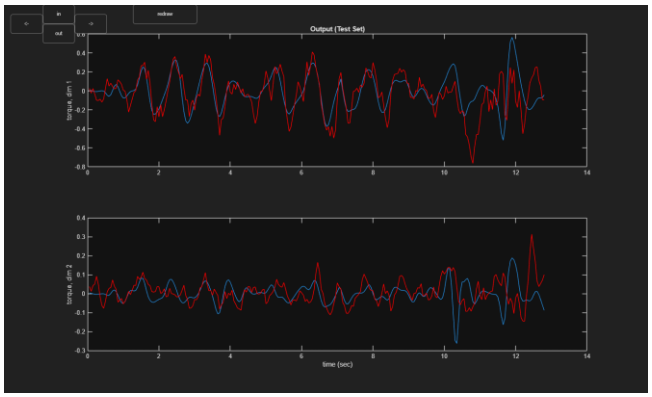
The model closely fits fold 1 but generalizes poorly to fold 10. It estimates weights for 960 neural features plus an intercept using only 1,194 training samples. Correlated neural features and overlapping history bins can make the regression sensitive to training-specific variation. The large training/test gap is consistent with overfitting. Negative elbow test FVAF means its squared prediction error exceeds the error from predicting the test set's mean elbow torque.

How do the training and test predictions compare?

The training predictions closely follow the actual shoulder and elbow traces. Test predictions have larger amplitude errors and missed peaks, especially for elbow torque. The initial displayed interval illustrates this difference, while FVAF summarizes the full evaluated fold.



Training



Test

Figure 1. One-fold torque predictions. Blue is actual and red is predicted; shoulder is above elbow. Original MATLAB displays.

Task 2 Training set effects

I increased the training set from folds 1 through 1–9 while keeping test fold 10 fixed. All other model settings stayed unchanged.

Folds	Train shoulder	Test shoulder	Train elbow	Test elbow
1	.9781	.2607	.9400	−.6840
2	.9260	.7583	.8038	.4114
3	.9075	.7872	.7566	.4767
4	.8910	.7869	.7090	.5087
5	.8505	.7777	.6426	.5125
6	.8350	.7981	.6215	.4962
7	.8340	.7981	.6161	.4968
8	.8068	.8079	.5740	.4971
9	.8083	.8139	.5737	.5074

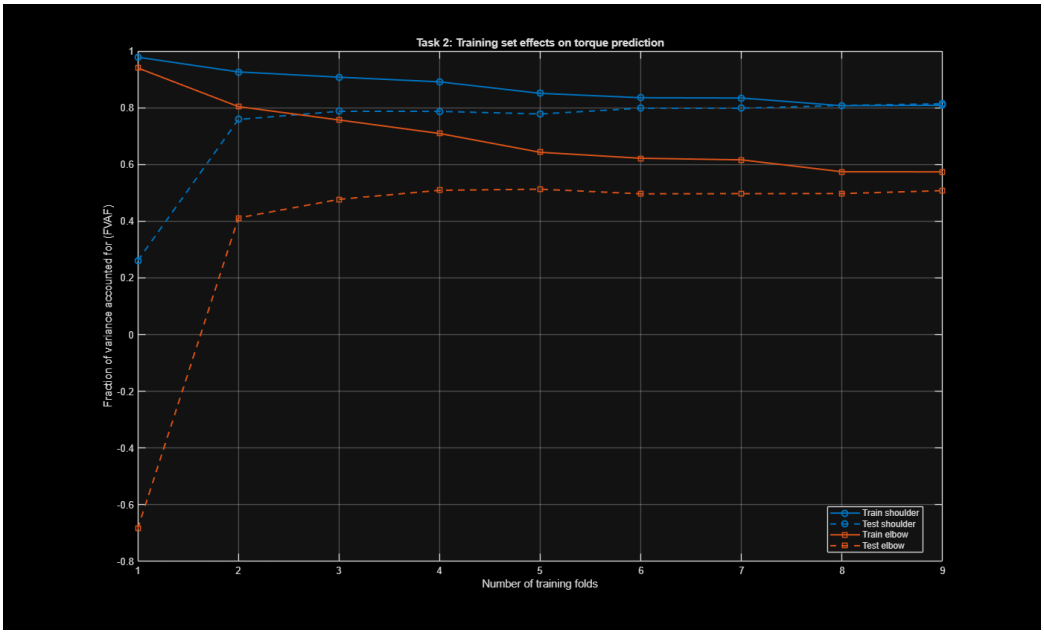


Figure 2. Training and test torque FVAF as training-fold count increases.

What changes from one to two folds and why?

Training FVAF decreases from 0.9781 to 0.9260 for shoulder and from 0.9400 to 0.8038 for elbow. The same model must fit more varied samples. Test FVAF improves sharply, from 0.2607 to 0.7583 for shoulder and from −0.6840 to 0.4114 for elbow. More training data reduces the sensitivity to one fold and improves generalization.

How many training folds would I choose?

I would use four folds as an efficiency compromise. Test shoulder FVAF is 0.7869 and elbow FVAF is 0.5087, already close to the nine-fold values. More folds give smaller, uneven gains. Nine folds gives the highest shoulder test FVAF, while five gives the highest elbow test FVAF. The choice depends on which output matters, available data, and computation. These results apply to this fold ordering and test set.

Task 3 Comparing prediction targets

I trained torque, Cartesian wrist position (X), and Cartesian wrist velocity (dX) predictors using identical inputs and folds and velocity. For torque, dimensions 1 and 2 are shoulder and elbow; for position and velocity, they are X and Y.

Target	Train dim 1	Test dim 1	Train dim 2	Test dim 2
Torque	.9260	.7583	.8038	.4114
Position	.8607	.4095	.8914	.2907
Velocity	.8885	.5497	.9097	.6560

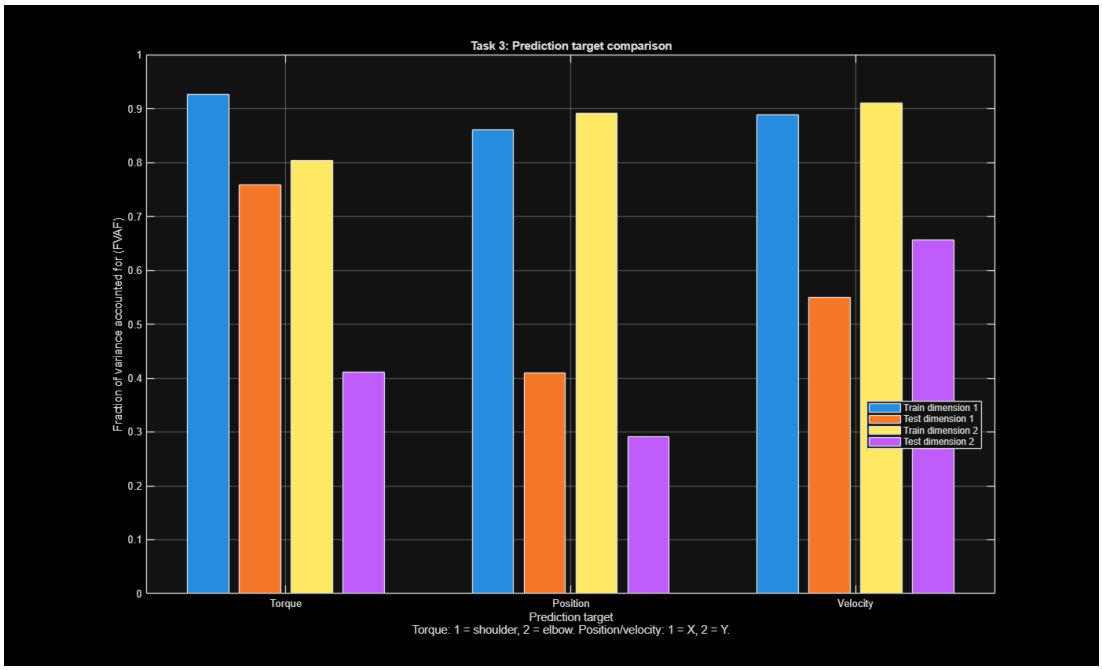


Figure 3. Grouped FVAF comparison for the three prediction targets.

How does Cartesian position compare with torque?

Both models follow their training trajectories fairly closely. Position generalizes less well: its test FVAF values are 0.4095 and 0.2907, compared with 0.7583 and 0.4114 for torque. Test position traces show offsets and missed changes, including the elevated actual Y position near samples 210–230. Torque captures many test shoulder oscillations, but elbow prediction misses several large excursions.

How does the additional predictor compare?

Velocity gives test FVAF values of 0.5497 in X and 0.6560 in Y. Its test traces capture many oscillations but miss or underestimate some sharp peaks. The unweighted mean of its two test FVAF values is 0.6028, compared with 0.5849 for torque and 0.3501 for position. Velocity has the highest mean, while shoulder torque has the highest individual test FVAF.

Conclusion from this data:

Prediction quality depends on the output variable and dimension. In this split, velocity and torque are predicted better on average than position. This does not establish that individual neurons exclusively encode one variable. The conclusion is limited to this linear model, its settings, and the selected folds.

Task 3 Visual comparisons

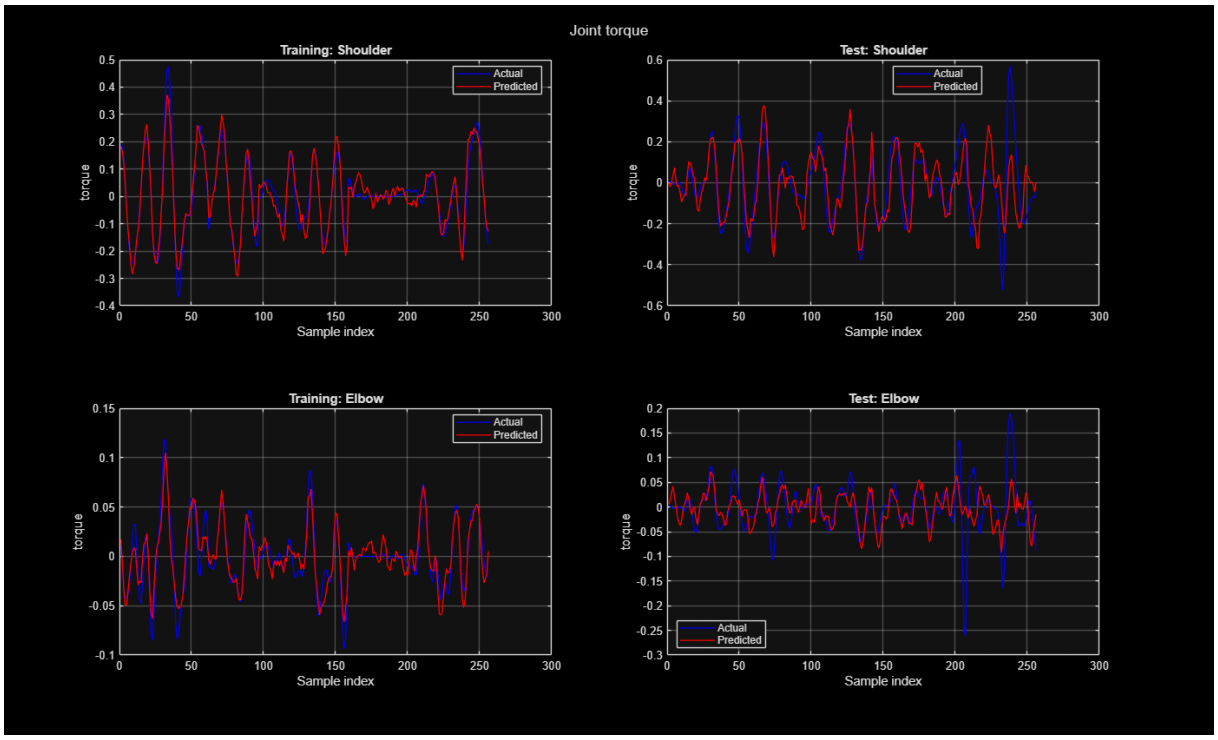


Figure 4. Torque predictions. Training is on the left and testing is on the right. Blue is actual; red is predicted.

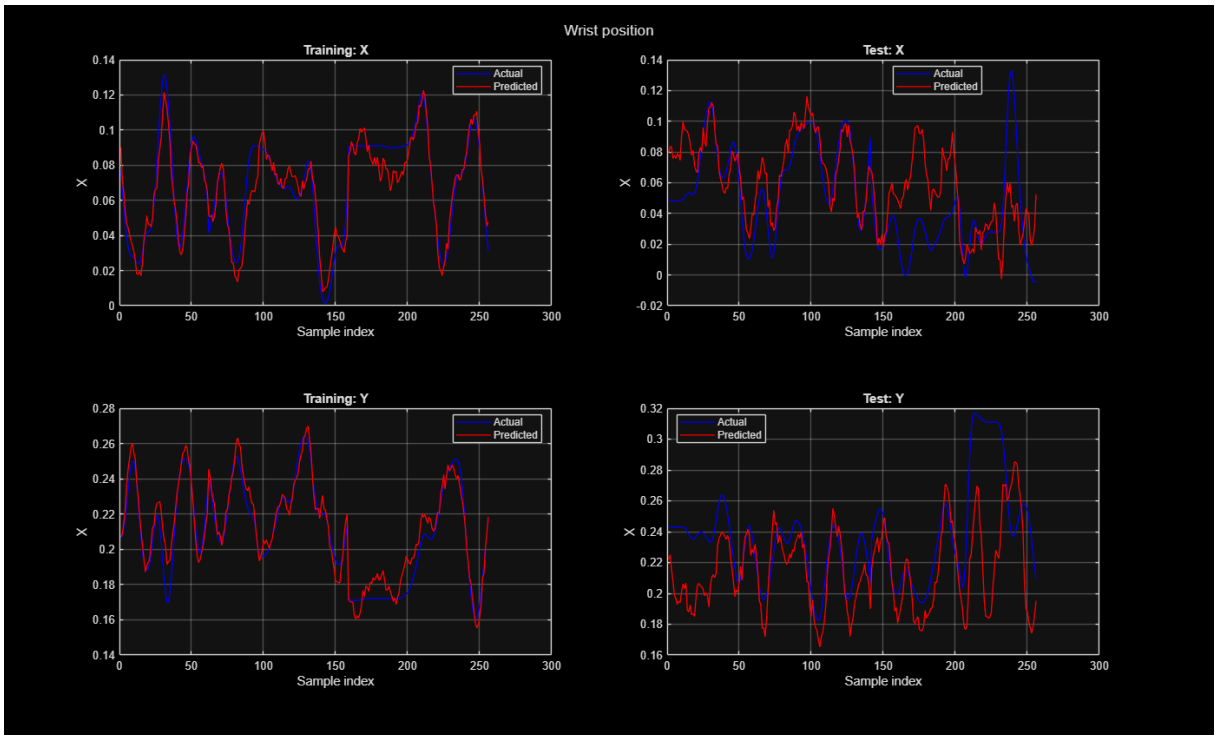


Figure 5. Position predictions. The lower row is Y, as labeled in the subplot titles. The original vertical labels use X for the overall target.

Task 3 Velocity and joint motion

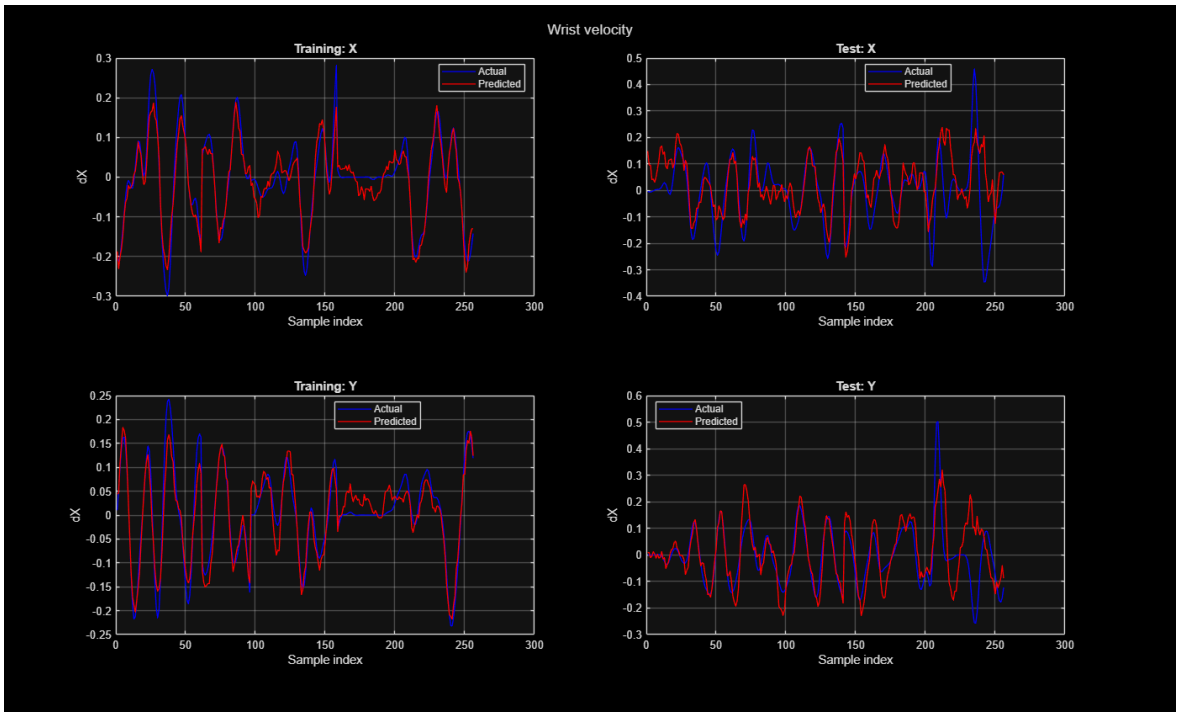


Figure 6. Velocity predictions for X and Y. Training is on the left and testing is on the right. The original vertical labels use dX for the overall target.

Task 3.1 Predicting joint motion

I changed the target to joint angle (theta), trained on folds 1–2, and animated both training and test predictions. Blue represented the actual arm and red represented the predicted arm.

Joint angle	Training FVAF	Test FVAF
Shoulder	0.9097	0.3081
Elbow	0.8656	0.2263

What did the animations show?

During training playback, the red and blue arms overlapped much more often and the prediction appeared less jittery than during testing. Test predictions sometimes pointed in a very different direction. For example, the actual arm could point near 2 o'clock while the prediction pointed near 5 o'clock. These mismatches persisted for about one or two seconds of visually observed playback.

The animation agrees with the FVAF results: the model reproduces familiar training movements much better than unseen test movements. The large gap is consistent with overfitting and limited generalization.

Task 4 Principal component preprocessing

I predicted torque from training folds 1–2 and tested on fold 10, with proprioception off and all 20 history bins. I tested 1–50 PCA components individually, then 100–950 in steps of 50, plus 960 components: 69 configurations total.

Components	Train shoulder	Test shoulder	Train elbow	Test elbow
No PCA	.9260	.7583	.8038	.4114
1	.0025	.0147	.0415	.0560
150	.8820	.8005	.6520	.4803
250	.8875	.7972	.6766	.4872
960	.9260	.7583	.8038	.4114

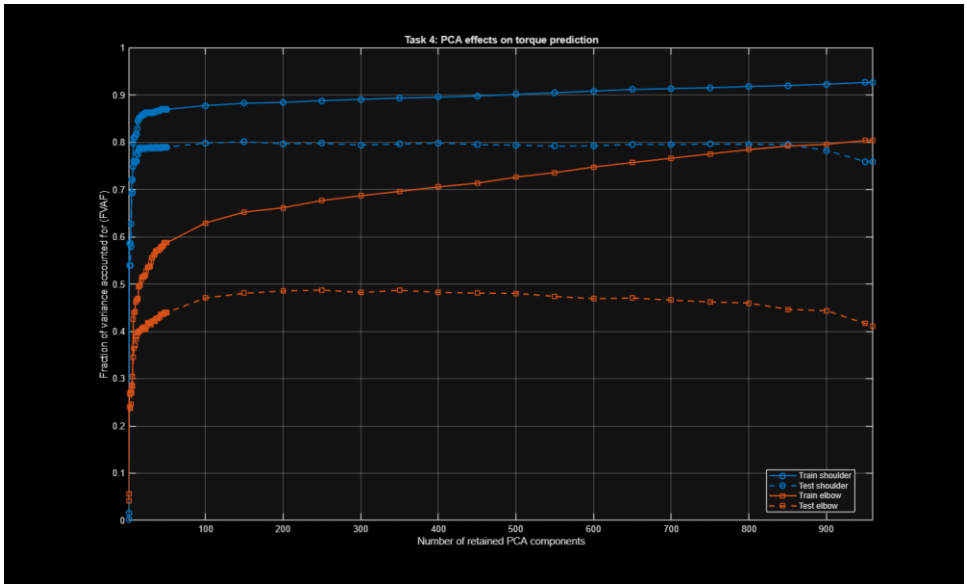


Figure 7. Torque FVAF across the sampled PCA component counts.

How does strong compression compare with no compression?

With one component, training and test FVAF are near zero for both joints, far below the uncompressed model. This is underfitting: too much information useful for predicting torque has been discarded. PCA retains input variance, which does not necessarily correspond to information relevant to the target.

What happens as components increase and where are the maxima?

Training FVAF increases as more components are retained. Test performance initially improves, then levels off and generally declines. The highest sampled shoulder test FVAF is 0.800494 at 150 components. Elbow test FVAF peaks at 0.487182 with 250 components. These are sampled maxima, not precisely established optimal component counts.

Why is there a maximum and how does it compare with no PCA?

Intermediate compression balances information loss against model flexibility. Adding components initially restores useful information, but later components can improve the training fit without helping generalization. Both sampled peaks exceed the uncompressed test scores of 0.7583 and 0.4114. Retaining all 960 components reproduces the uncompressed results, providing a consistency check.

Task 5 Proprioceptive inputs

I compared torque prediction with proprioception off and on across one through nine training folds, keeping test fold 10 fixed and PCA off. Enabling proprioception adds shoulder and elbow angles and angular velocities: four extra inputs, changing the total from 961 to 965 columns including the intercept.

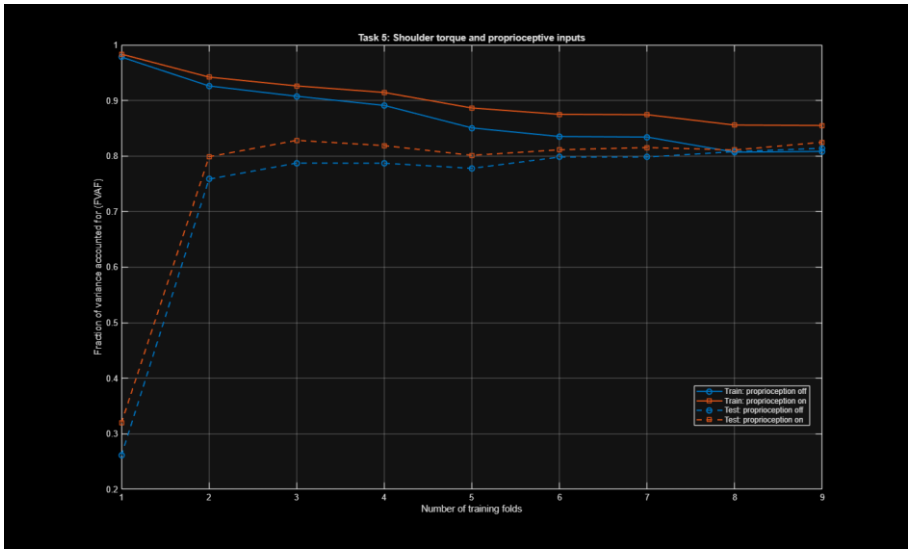


Figure 8. Shoulder torque with proprioception off and on.

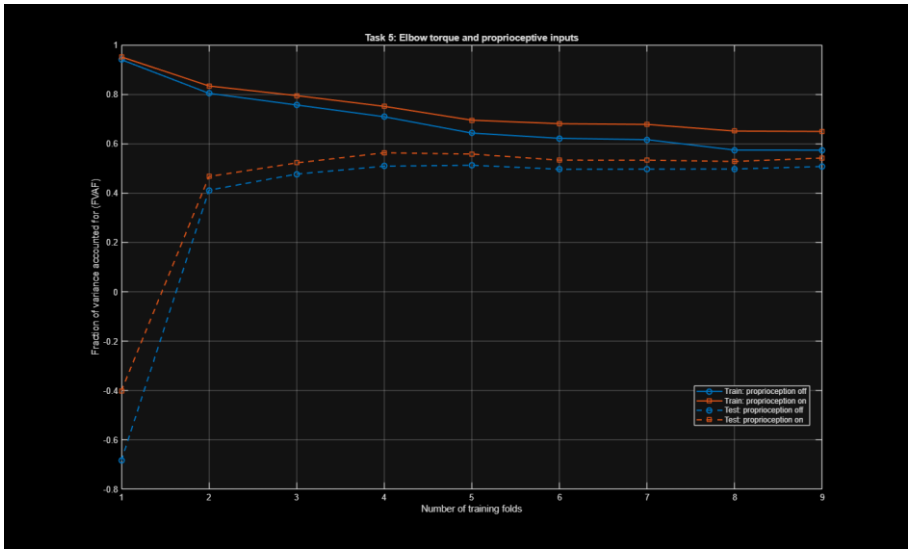


Figure 9. Elbow torque with proprioception off and on.

How does proprioception affect training and test performance and why?

Training and test FVAF improve for both joints at every training size tested. With two training folds, shoulder test FVAF increases from 0.7583 to 0.7986 and elbow test FVAF from 0.4114 to 0.4669. The highest sampled scores with proprioception are 0.8281 for shoulder at three folds and 0.5629 for elbow at four folds.

The added joint-state inputs provide information useful for predicting torque beyond neural activity alone. More inputs also give the regression more flexibility, so better training fit alone would not prove better generalization. The improvements on held-out fold 10 provide that additional evidence for this split. Proprioception does not solve insufficient training data: one-fold elbow test FVAF remains negative at −0.4016. The model also requires measured joint-state inputs during testing.

Task 6 Varying neural history

I varied neural history from 1 to 20 consecutive bins, each covering 50 ms. Torque prediction used training folds 1–2 and test fold 10, with PCA and proprioception off. Each additional bin adds 48 neural features, while sample counts remain fixed.

History	Train shoulder	Test shoulder	Train elbow	Test elbow
50 ms	.5662	.4956	.3332	.2358
500 ms	.8932	.7947	.7263	.4629
650 ms	.9058	.7865	.7507	.4714
1000 ms	.9260	.7583	.8038	.4114

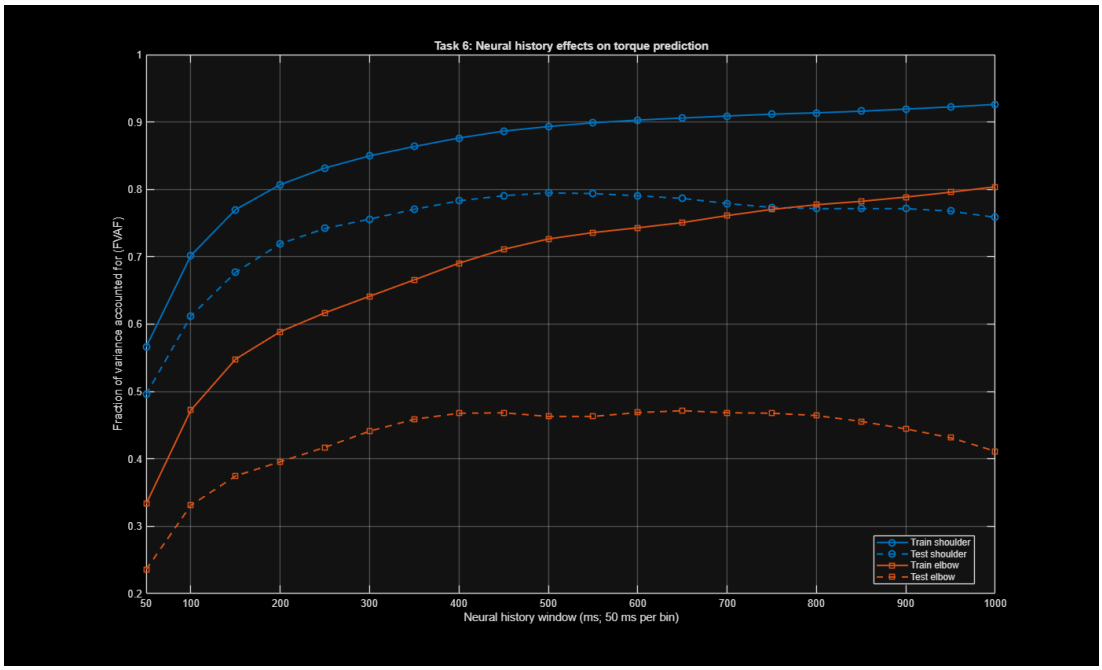


Figure 10. Torque FVAF as the neural-history window increases.

How does performance change as history increases?

Training FVAF increases for both joints. Test FVAF initially improves, showing that previous neural activity contributes useful information for predicting current torque. With longer histories, test performance levels off and then generally declines, even though training fit continues improving. Adding more features increases flexibility and can fit training-specific variation. This pattern is consistent with increased overfitting.

At what history length does test performance reach a maximum?

Shoulder test FVAF peaks at 0.794737 with 10 bins, a 500 ms window. Elbow test FVAF peaks at 0.471430 with 13 bins, a 650 ms window. A 500–650 ms window is a reasonable range for this experiment, although the elbow curve is fairly flat around its maximum. The full 20-bin history reproduces the two-fold uncompressed baseline and performs worse than these peaks.

Window duration counts the widths of all selected bins. For example, 10 bins cover 500 ms, with bin delays labeled 0 through 450 ms. These findings concern prediction under this model and split, rather than a uniquely determined physiological delay.

Task 7 Optimizing the model

The regression weights W are parameters learned from data. Hyperparameters are choices made before fitting, such as PCA component count, history length, and whether to include proprioception.

Why is changing one hyperparameter at a time insufficient?

Hyperparameters interact, so the best value of one setting can depend on the others. In Task 4, I selected PCA counts while using 20 history bins. In Task 6, I compared history lengths with PCA disabled. Shortening history changes the number and identity of the inputs, which also changes the PCA representation. Combining the separate best settings therefore does not guarantee the best overall model. Training size and proprioception can also change which configuration generalizes best.

What technique could optimize the hyperparameters together?

Grid search with cross-validation can evaluate combinations of settings. I would first fix torque as the prediction objective and define the selection score, such as mean shoulder/elbow validation FVAF. Changing the output target changes the prediction problem, so I would compare target types separately rather than treating them as interchangeable ways to improve torque prediction.

1. Define candidate combinations of history length, PCA count, and proprioception. Restrict PCA counts to the available input dimension.
2. Evaluate each combination across training/validation splits made from the existing data folds. Avoid randomly mixing adjacent time samples across sets.
3. Fit standardization, PCA, and regression using only the training portion of each split. Apply the fitted transformations to its validation portion.
4. Select the combination with the best average validation score, considering variability and model complexity when scores are similar.
5. Retrain the selected model on the development data and evaluate it once on an untouched test set. Nested cross-validation can separate selection from evaluation when using multiple outer splits.

Training size can be studied with controlled learning-curve comparisons within the development data. Once a configuration is selected, I would normally fit it using all available development training data.